



MAGNETIC FIELD



Introduction:

A magnetic field is a fundamental concept in physics that describes the region around a magnet or a current-carrying conductor where magnetic forces are exerted on other magnets, conductors, or charged particles.

Permanent magnets, such as iron magnets, have their own intrinsic magnetic fields due to the alignment of their magnetic domains.

Magnetic fields have numerous practical applications, including:

- Electric motors
- Transformers
- MRI machines (Magnetic Resonance Imaging)
- Magnetic levitation trains
- Magnetic compasses
- Magnetic data storage (hard disk drives)

Magnetic Fields:

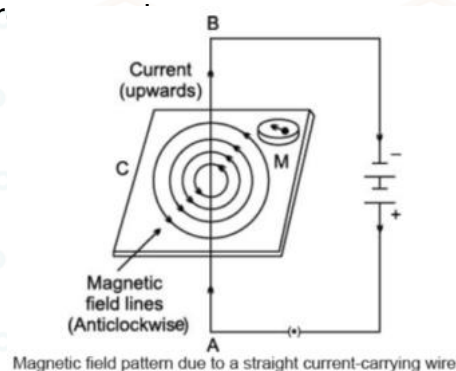
Magnetic field can be defined as "An area around a permanent magnet or a current carrying conductor, where they generate a magnetic force".

Magnetic fields can be generated through two primary methods:

- From permanent magnets.
- From current carrying conductors, also known as elect

Magnetic Force on a Current Carrying Conductor:

When we place a current-carrying conductor in a uniform External magnetic field, the interaction between the magnetic field produced by the conductor and the external magnetic field gives rise to external force, denoted as F , acting on the conductor, as illustrated in figure.



Factors on which the force acting on current carrying conductor in a magnetic field depend:

The force acting on a current-carrying conductor depends on several factors, such as length, current, and the strength of the external magnetic field.

The force (F) is directly proportional to the length of the conductor (L) that lies within the external magnetic field:

$$F \propto L$$

The force is also directly proportional to the current (I) passing through the conductor.

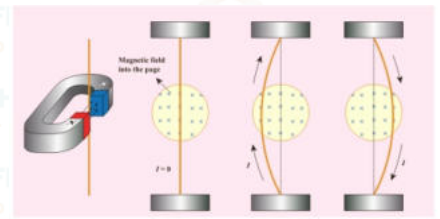


Figure 5.7 Deflection of current carrying wire in magnetic field

Similarly, the force is directly proportional to the strength of the external magnetic field (B).

$$F \propto B$$

Combining these factors, we will obtain the following relationship:

$$F \propto BIL$$

Here, "k" is the proportionality constant, and in the SI unit system, its value is equal to 1.

Therefore, we can express the force as:

$$F = BIL$$

If the magnetic field is not perpendicular to the wire, Above equation can be modified to:

$$F = I(L \times B)$$

where 'x' denotes the vector cross product.

The magnitude of the force (F) is determined by:

$$F = BIL \sin \theta$$

Where, θ represents the angle between the direction of the conductor's length (L) and the external magnetic field (B).

Maximum Force:

The deflecting force reaches its maximum when the angle (θ) between the length of the conductor and the external magnetic field is $= 90^\circ$ degrees, assuming other factors remain constant.

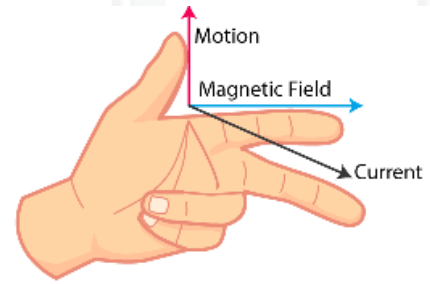
Minimum Force:

Conversely, the deflecting force is minimized when $= 0$ degrees, signifying that the magnetic field and conductor are aligned.

Since magnetic force is a vector quantity, its direction can be determined using Fleming's lefthand rule, which provides a straight forward way to establish the direction of the force in relation to the current and magnetic field.

Thrust or Fleming's left-hand rule: Motion Motion:

Fleming's left-hand rule is used to determine the direction of force acting on a current carrying wire placed in a magnetic field, also to identify the direction of the forces in an electric motor in order to apply Fleming's left-hand rule, follow these steps: Extends our fore finger, thumb, and Middle finger such figure 18. 3 Fleming's left-hand rule that they are mutually perpendicular as shown in figure.



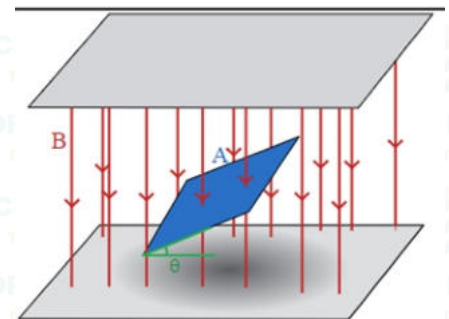
Magnetic Flux:

Magnetic flux is defined as the number of magnetic lines of force passing through an area held perpendicular to it.

It is calculated by taking the dot product between the magnetic field and area vector.

$$\Phi_B = B \cdot A$$

$$\Phi_B = BA \cos \theta$$



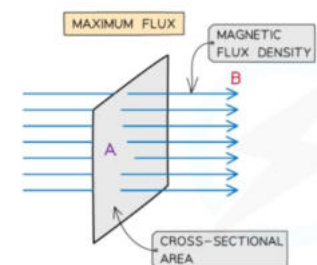
Where B is uniform magnetic field

A is an area vector whose magnitude is equal to the surface element and directed perpendicular to the given surface element.

Maximum Magnetic Flux:

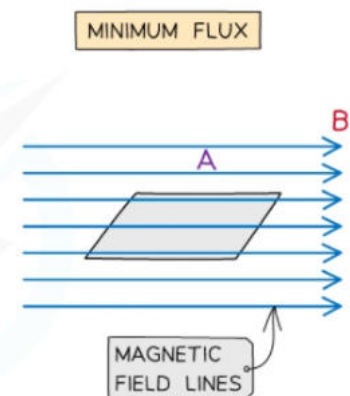
The magnetic flux will be maximum when the angle between magnetic field B and area vector A is zero,

i.e. $\theta = 0^\circ$ surface will be perpendicular to the magnetic lines. This case is shown in figure,



Minimum Magnetic Flux:

Similarly when the angle between area vector and magnetic field is $\theta = 90^\circ$ then lines of force do not pass through the surface and magnetic flux will be minimum as shown in figure.



Unit of Magnetic Flux:

The SI unit of magnetic flux is Weber (Wb) = $1\text{Wb} = \frac{Nm}{A}$

Magnetic Flux Density:

The magnetic flux density at a specific point in space is the force experienced per unit length by along straight conductor carrying unit current, placed perpendicular to the field at that particular location.

$$B = \frac{F}{IL}$$

Unit of Magnetic Flux Density:

The SI unit of magnetic flux density is Tesla (T), named after the Serbian-American inventor Nikola Tesla.

One Tesla: is defined as follows, If a conductor having length 1 m and carries a current of 1A placed perpendicularly the magnetic field experience a force of one Newton then magnetic flux density will be 1 Tesla.

In essence, the Tesla provides a standardized measure of the strength of a magnetic field, with larger values indicating stronger magnetic fields and smaller values representing weaker fields.

Amperes Law:

It is known that an electric current flowing through a wire creates a magnetic field in its vicinity. When we envision a closed circular path with the wire positioned at its center, the Magnetic flux density (B) within this 4 circular region changes, depending on both the current (I) and the distance (r) from the wire as shown in figure

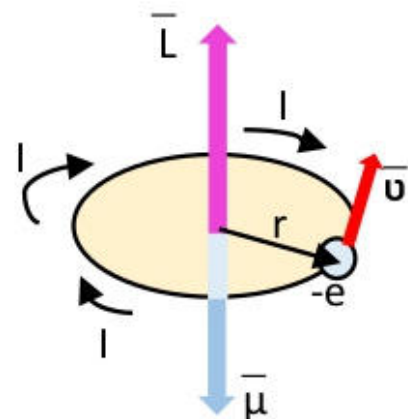
$$B \propto 2L \dots\dots\dots i$$

Or

$$B \propto \frac{1}{r} \dots\dots\dots ii$$

Combing (i) and (ii), we get

$$B \propto \frac{2L}{r}$$
$$B \propto \frac{\mu_0 I}{2\pi r}$$



Where μ_0 is permeability of free space = $4 \times 10^{-7} \text{Wb/A.m}$

Now consider an Amperian loop in form of a closed path around the current carrying wire as shown in figure 18.7. For any path element AL we can write

$$B \cdot AL = \mu_0 I$$

Since angle between B and L is zero so $B \cdot AL = BAL = \mu_0 I$

Now adding up over whole closed path

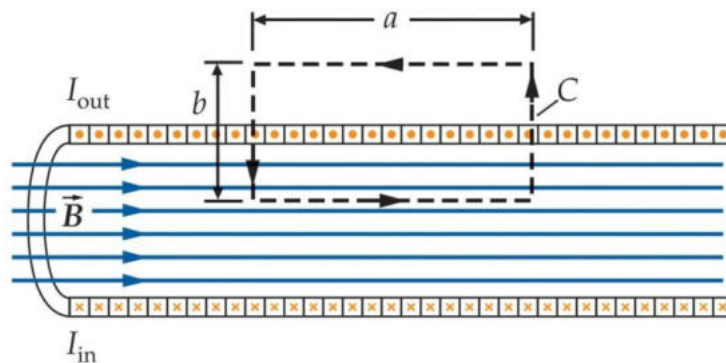
$$\oint B \cdot dL = \mu_0 I$$

Hence Ampere's law can be defined as the integral of the magnetic field (B) around a closed loop is directly proportional to the total electric current (I) passing through the loop

Application of Ampere's Law:

(i) Solenoid:

A solenoid is a long, tightly wound coil of wire, and often used to generate a uniform magnetic field within the coil's interior. It consists of many closely spaced turns of wire, and when an electric current flows through it, it creates a magnetic field that is similar to the field produced by a bar magnet. To determine the magnitude of B of a solenoid let us consider an Amperian loop abcd with lengths l_{ab} , l_{bc} , l_{cd} and l_{da} shown in figure.



Then from Ampere's law

$$\oint B \cdot dL = \mu_0 I$$

The left hand side of equation is

$$\oint B \cdot dL = -B \cdot l_{ab} + B \cdot l_{bc} + B \cdot l_{cd} + B \cdot l_{da}$$

As the element at all lengths along bc and da are perpendicular to the magnetic field so,

$$B \cdot l_{bc} \text{ and } B \cdot l_{da} = 0$$

Moreover, the magnetic field intensity outside of solenoid ≈ 0 , so $B_{\text{cd}} = 0$

Hence the magnetic field due to solenoid is given by

$$\Sigma B \cdot \Delta l = B \cdot L_{\text{ab}} = \mu_0 I$$

For N number of turns on the solenoid

$$\Sigma B \cdot \Delta l = B \cdot L_{\text{ab}} = \mu_0 NI$$

The number of turns per unit length is given by $n = \frac{N}{L_{\text{ab}}}$

$$B = \mu_0 n I$$

Hence it is shown that B is independent of the position within the solenoid which shows that the field is uniform in a solenoid.

(ii) Toroid:

A toroid can be considered as a solenoid that is bent into the shape of a circle. It also generates a magnetic field within its interior. The magnetic field lines inside a toroid are generally circular and concentrated within the coil.

Suppose a toroid consists of NNN closely packed turns of copper wire and carry a current I along it.

Imagine a hollow circular ring wound with NNN number of current-carrying wire. In this case, within the toroid, point P is our point of interest, and we will determine the magnetic field.

At this location as B, inside the toroid, we construct an Amperian loop forming a circle that passes through point P. This arrangement gives rise to concentric circles within the toroid.

Because of the symmetrical nature of the magnetic field, the magnitude of the field at all points along the circle is uniform, and the field direction is tangential to the circle.

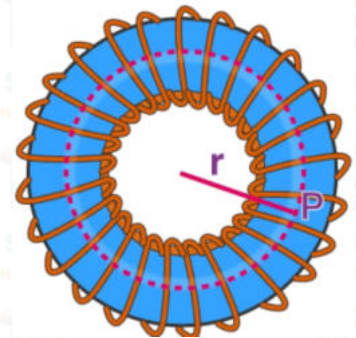
Therefore,

$$\Sigma B \cdot \Delta l = \Sigma B \cdot \Delta l \cos(0) = B \Sigma \Delta l = (2\pi r)$$

Now, we can apply Ampere's law:

$$2\pi r = \mu_0 NI$$

$$B = \frac{\mu_0 NI}{2\pi r}$$



This is the expression for the magnetic field strength inside a toroid. It is inversely proportional to the radius and directly proportional to the number of turns per unit length NN and the current I flowing through the turns.

Force on a charged particle in a uniform magnetic field:

When a charged particle moves through a uniform magnetic field, it experiences a force known as the magnetic force. This phenomenon is described by the Lorentz force equation, which can be written as:

$$F = (v \times B)$$

Charge to mass ratio of Electron:

Measuring the charge-to-mass ratio (e/m_e) of an electron is a classic experiment in the field of electromagnetism and particle physics. This experiment, known as the "e/m experiment," involves applying both a magnetic field and an electric field to a beam of electrons." Here's a step-by-step method to measure the e/m_e of an electron:

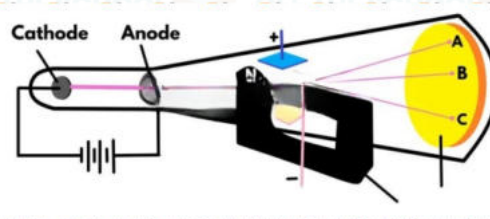
Apparatus and Materials:

1. **Cathode Ray Tube (CRT):** This is a vacuum tube that produces a beam of electrons. It consists of an electron gun, focus, in deflection plates, and a fluorescent screen.
2. **Magnetic Field Source:** We'll need a strong uniform magnetic field source, such as a Helmholtz coil or a solenoid.
3. **Voltage Source:** A variable voltage source to create an electric field.
4. **Fluorescent Screen:** A screen coated with a phosphorescent material to visualize the electron path.
5. **Rulers and Measurement Devices:** To measure the radius of the electron beam's circular path and the electric potential applied.

Experimental Procedure:

1. Setup:

Set up the CRT in a vacuum chamber to ensure the electron beam doesn't interact with air molecules.



Position the Helmholtz coil (or solenoid) around the CRT, providing a uniform magnetic field parallel to the beam path.

Connect the voltage source to the focusing and deflection plates to create an electric field perpendicular to the magnetic field.

2. Calibration:

Calibrate the magnetic field by measuring its strength at the location of the electron beam. We can do this using a magnetic field strength meter.

3. Electron Beam Production:

Apply a high voltage across the cathode and anode of the CRT to generate a beam of electrons from the cathode (electron gun).

Use the focusing and deflection plates to control and direct the electron beam.

4. Observe the Electron Beam:

Turn off the magnetic field and electric field to observe the electron beam's straight-line path on the fluorescent screen.

5. Apply the Magnetic Field:

Turn on the magnetic field, which causes the electron beam to bend in a circular path due to the Lorentz force (the interaction between the magnetic field and the moving electrons).

6. Measure the Radius(r):

Measure the radius of the circular path formed by the electron beam on the screen. Ensure the screen is marked with a scale for accurate measurement.

7. Apply the Electric Field:

Apply a known electric potential (voltage) across the focusing and deflection plates perpendicular to the magnetic field. This will introduce an electric force on the electrons.

8. Adjust the Voltage:

Adjust the voltage until the electron beam returns to its original, undeviated straight-line path.

9. Measure the Electric Potential (V):

Measure the voltage applied across the plates.

Calculations:

An electron is released from the cathode and it gains speed while passing through the potential difference V , hence gaining kinetic energy of eV . This electron kinetic energy is calculated by:

$$\frac{1}{2}mv^2 = eV$$

The magnetic field produced by Helmholtz coils is perpendicular to this velocity, and produces a magnetic force which is transverse to both v and B : this provides the centripetal force makes an electron move along the circular trajectory; the radius of this trajectory r can be found from:

$$mv^2 = qvB$$

From this equation, we obtain the expression for the charge-to-mass ratio of the electron, we can calculate the e/m_e ratio of the electrons which was found to be $1.7588 \times 10^{11} \text{ C/kg}$

Measuring Instruments:

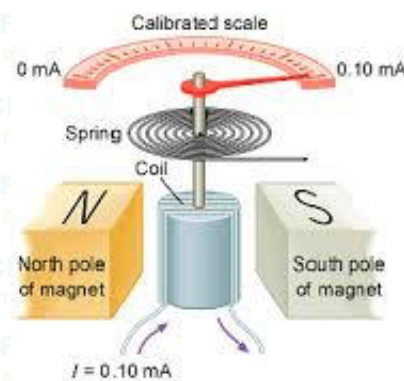
Galvanometer:

A Galvano meter is an instrument used to detect and measure electric currents. It is a highly sensitive electromagnetic apparatus capable of measuring even very small currents, such as those on the order of a few micro amperes.



Principle:

It works based on the principle of electromagnetic induction. When a coil carrying an electric current is positioned within an external magnetic field, it undergoes magnetic torque. The degree of deflection observed in the coil, caused by this magnetic torque, is directly proportional to the current's magnitude flowing through the coil.



Construction:

1. Coil:

The key component of a Galvano meter is a coil of wire (usually wound around a soft iron core) suspended within a magnetic field. The coil is mounted on a spindle so that it can rotate freely.

2. Magnet:

A permanent magnet or an electromagnet is placed around the coil. The magnetic field lines from the magnet pass through the coil.

3. Spring:

A delicate torsion spring is attached to the coil, providing a restoring torque when the coil is deflected.

4. Pointer:

A thin pointer or needle is attached to the coil, allowing for the measurement of the deflection.

Working:

Consider a rectangular coil consisting of N turns, each with a current I flowing through them and a cross-sectional area A . When this coil is situated within a uniform radial magnetic field B , it undergoes a torque as shown in the figure.

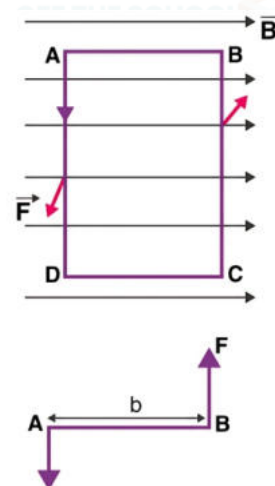
Let's examine a single turn $ABCD$ of the rectangular coil, characterized by a length L and breadth b as shown in the figure. This turn is suspended within a magnetic field with strength of B , arranged so that the coil's plane is parallel to the magnetic field lines. As sides AB and DC are aligned parallel to the magnetic field, they do not experience any discernible force due to the magnetic field. However, sides AD and BC , which are perpendicular to the field's direction, encounter an effective force denoted as F , given by:

$$F = BIL$$

By employing Fleming's left-hand rule, we recognize that the forces acting on sides AD and BC are oriented in opposite directions each other. When a pair of equal and opposite forces, denoted as F and collectively referred to as a couple, act on the coil, they generate a torque. This torque induces a deflection in the coil.

The torque (τ) is calculated as the product of the force (F) and the perpendicular distance (b) between these forces:

$$\tau = F \times b$$



Substituting the known value of F, we have:

Torque(t) acting on a single-loop ABCD of the coil

$$T = BIL \times b$$

Where $L \times b$ represents the area (A) of the coil. Consequently, the torque acting on a coil with n turns is given by:

$$T = NIAB$$

This magnetic torque causes the coil to rotate, leading to the twisting of the phosphor bronze strip. Simultaneously, the spring (S) attached to the coil exerts a counter-torque, known as the restoring torque ($k\theta$), resulting in a stable angular deflection

Under equilibrium conditions, we find:

$$k\theta = NIAB$$

Here, k is termed the torsional constant of the spring, representing the restoring couple per unit twist. The deflection or twist (θ) is quantified as the reading displayed on a scale by a pointer connected to the suspension wire.

$$\theta = \left(\frac{NAB}{K} \right) I$$

$$\theta \propto I$$

Therefore,

the quantity $\frac{NAB}{K}$ is a constant for a given galvanometer.

Therefore, it is evident that the deflection observed in the galvanometer is directly proportional to the current flowing through it.

Conversion of Galvanometer to Ammeter:

To convert a galvanometer into an ammeter, a low-resistor known as a shunt is connected in parallel with the galvanometer.

The selection of an appropriate shunt is based on the desired current range of the ammeter.



From the circuit as shown in figure:

- R_g represents the resistance of the galvanometer.
- G denotes the Galvanometer coil.
- I stands for the total current flowing through the circuit.
- I_g represents the total current passing through the galvanometer, which corresponds to the full-scale reading.
- R_s represents the value of the shunt resistance.

When the current I_g flows through the Galvanometer, the current passing through the shunt is determined by:

$$I_s = I - I_g$$

The voltages across both the galvanometer and shunt are R_s equal due to their parallel connection.

Hence, we can establish the following equation:

$$R_g I_g = (I - I_g) \times R_s$$

$$R_s = \left(\frac{R_g I_g}{(I - I_g)} \right)$$

Conversion of Galvanometer to Voltmeter

A galvanometer is converted into a voltmeter by connecting it in series with high resistance as shown in the figure. A suitable high resistance is chosen depending on the range of the voltmeter.

- R_g stands for the resistance of the galvanometer.
- R_x represents a high resistance component called series resistance.
- G denotes the galvanometer coil.
- I is the total current flowing through the circuit.
- I_g signifies the total current passing through the galvanometer, corresponding to a full-scale deflection.
- V represents the voltage drop across the series connection of the galvanometer and the high resistance.

When the current I_g flows through the series combination of the galvanometer and the high resistance R , the voltage drop across the branch ab can be expressed as:

$$V = V_g + V_x$$

Current in the series combination remains conserved therefore:

$$I_g = I_k$$

$$V = I_g R_g + I_g R_x$$

$$I_g R_x = V - I_g R_g$$

$$R_x = \frac{V}{I_g} - \frac{I_g R_g}{I_g}$$

$$R_x = \frac{V}{I_g} - R_g$$

This equation can be used to determine the value of R_x .